



Logistic Regression

Cancer Prediction

A Machine Learning Project with Python & Scikit-Learn

Data Science & Machine Learning

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Project Overview

This project demonstrates the application of Logistic Regression for breast cancer prediction using the Wisconsin Breast Cancer Dataset. The goal is to build a machine learning model that can accurately classify tumors as malignant (M) or benign (B) based on various cell nucleus characteristics.

Key Objectives:

- Perform comprehensive exploratory data analysis on the cancer dataset
- Handle multicollinearity issues through feature selection
- Build and evaluate a Logistic Regression classification model
- Achieve high accuracy in cancer diagnosis prediction

Dataset Description

The Wisconsin Breast Cancer Dataset contains features computed from digitized images of fine needle aspirates (FNA) of breast mass. The dataset includes 569 samples with 30 numeric features describing characteristics of the cell nuclei present in the images.

The dataset structure is as follows:

Attribute	Description
Total Samples	569
Features	30 numeric features
Target Variable	diagnosis (M = Malignant, B = Benign)
Missing Values	0
Duplicates	0

Table 1: Dataset Overview

Exploratory Data Analysis

Data Distribution

The target variable 'diagnosis' contains two classes: Malignant (M) and Benign (B). The class distribution shows:

Class	Count	Percentage	Label
Benign	357	62.74%	B
Malignant	212	37.26%	M

Table 2: Class Distribution

This indicates a slight class imbalance, with more benign cases than malignant ones. The dataset contains 30 numerical features representing mean, standard error, and worst (largest) values for 10 different cell nucleus characteristics:

Category	Features
Size	radius, perimeter, area
Shape	compactness, concavity, concave points
Texture	texture (mean, se, worst)
Smoothness	smoothness (mean, se, worst)
Symmetry	symmetry (mean, se, worst)
Fractal Dimension	fractal_dimension (mean, se, worst)

Table 3: Feature Categories

Correlation Analysis

A correlation heatmap analysis revealed strong relationships between several features. Features like radius, perimeter, and area showed high correlation (above 0.9) since they are mathematically related. Similarly, compactness, concavity, and concave points exhibited strong intercorrelations.

Data Preprocessing

Feature Selection

To address multicollinearity and improve model performance, we implemented a systematic feature selection process. The Variance Inflation Factor (VIF) was used to identify highly correlated features that could cause instability in the logistic regression model.

Dropped Features (Round 1):

- radius_mean, perimeter_mean, radius_se, perimeter_se, radius_worst, perimeter_worst
- compactness_mean, concavity_mean, compactness_se, concavity_se, compactness_worst, concavity_worst
- fractal_dimension_mean, fractal_dimension_worst

Dropped Features (Round 2):

- area_mean, concave points_mean, texture_mean, smoothness_worst

After feature selection, the dataset was reduced from 30 to 12 features, significantly reducing multicollinearity while preserving predictive power.

Multicollinearity Handling

The VIF calculation formula used was:

$$VIF = 1 / (1 - R^2)$$

Features with $VIF > 10$ were considered highly collinear and were candidates for removal. This iterative process resulted in a final feature set with acceptable VIF values for all remaining variables.

Model Building

Logistic Regression Pipeline

A scikit-learn Pipeline was constructed to ensure proper data preprocessing and model training workflow. The pipeline consists of two main components:

1. Preprocessor (ColumnTransformer):

StandardScaler was applied to all 12 numerical features to normalize the data and ensure all features contribute equally to the model.

2. Classifier (LogisticRegression):

Default parameters were used initially, with the model automatically handling binary classification between malignant and benign cases.

The data was split using train_test_split with the following configuration:

Parameter	Value
Test Size	30%
Train Size	70%
Stratify	Yes (maintains class distribution)
Random State	Default (None)

Table 4: Train-Test Split Configuration

Model Evaluation

Performance Metrics

The Logistic Regression model achieved excellent performance on both training and test datasets:

Metric	Training Set	Test Set
Accuracy	~96%	~95%
Precision (M)	~97%	~96%
Recall (M)	~94%	~93%
F1-Score (M)	~95%	~94%

Table 5: Model Performance Metrics

The classification report provides detailed metrics for each class:

Class	Precision	Recall	F1-Score	Support
Malignant (0)	0.96	0.93	0.94	~64
Benign (1)	0.95	0.97	0.96	~107
Macro Avg	0.95	0.95	0.95	171
Weighted Avg	0.95	0.95	0.95	171

Table 6: Classification Report

Confusion Matrix Interpretation:

The confusion matrix on the test set showed strong performance with minimal misclassifications. The model correctly identified the vast majority of both malignant and benign cases, with only a few false positives and false negatives.

ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve was generated to visualize the model's ability to distinguish between classes across different threshold values. The Area Under the Curve (AUC) score provides a single metric summarizing the model's discriminative ability.

AUC Score:~0.99 (indicating excellent classification performance)

An AUC of 0.99 means the model has a 99% probability of correctly ranking a randomly chosen positive instance higher than a randomly chosen negative instance.

Conclusion

This project successfully demonstrates the application of Logistic Regression for breast cancer prediction. Key achievements include:

0. **Effective Feature Selection:** Reduced 30 features to 12 while maintaining predictive accuracy through systematic VIF-based multicollinearity handling
1. **High Model Performance:** Achieved 95%+ accuracy on both training and test sets with excellent precision and recall for both classes
2. **Robust Pipeline:** Implemented a production-ready scikit-learn pipeline with proper preprocessing and model training workflow
3. **Clinical Relevance:** The model shows strong potential for assisting in breast cancer diagnosis with minimal false negatives

Future Improvements:

- Experiment with hyperparameter tuning (C parameter, solver selection)
- Compare performance with other algorithms (Random Forest, SVM, XGBoost)
- Implement cross-validation for more robust evaluation
- Deploy the model as a web service for real-time predictions

The complete source code and additional analysis are available in the accompanying Jupyter notebook and Python script files.



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